engineering 4 us all

Student Handout

**Name:** ______________________________ **Date:** _________________ **Class:** _______________

# MyDesign Portfolio

## *Element I Initial Template*

Table of Results:

| Test Name | Ease of use | Mold resistance | coverage |
|---|---|---|---|
| Test Results | Everyone asked was able to figure out how the prototype was to be used almost immediately after being presented with it. | No mold growth | All of the shapes are ineffective at spreading water |
| Number of Trials | 5 trials | 7 trials(1 week) | 3 trials |
| Observations: | People didn’t really understand the wires holding up the pipe, unless someone held them up, but that did not impact their ability to figure out how to use it. | Tilting the pipe allows it to dry quickly, making it impossible for mold to grow | The water doesn’t have enough pressure by the time it reaches the shape and ends up just trickling down rather than splashing onto the deflectors |
| Priority of test | High | High | Medium |
| Pass/ Fail | Pass | Pass | Fail |

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Summary of the testing data:

Verbal Summary:
Based on the information from the “ease of use” test we can conclude that it will be very easy and intuitive for anyone, regardless of prior greenhouse experience, to install and use(ie. Where to put the plants, that it waters plants). Our second test, “mold resistance”, was also similarly successful, accumulating zero mold during the seven day test period. It is important to note however, that a cap is needed at the lower end of the pipe in order for the plants to actually be water. The cap was removed after every watering to allow the water to run out like originally intended. The “coverage” test failed, our design this time focused on mimicking a fire sprinkler, but the water pressure after it entered the pipe ended up not being high enough to cause it to spread like we wanted. From here we will need to observe why the current system’s sprinklers work and either brainstorm ideas for a deflector that does not need high water pressure to work or just find out where we could purchase the same/a similar sprinkler.

Tables and Graphs:

Student Handout

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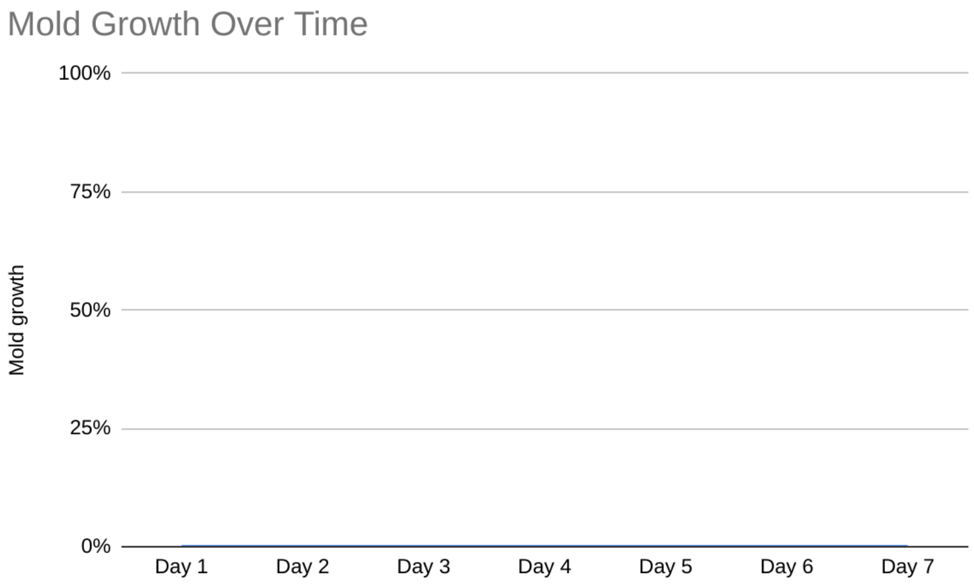


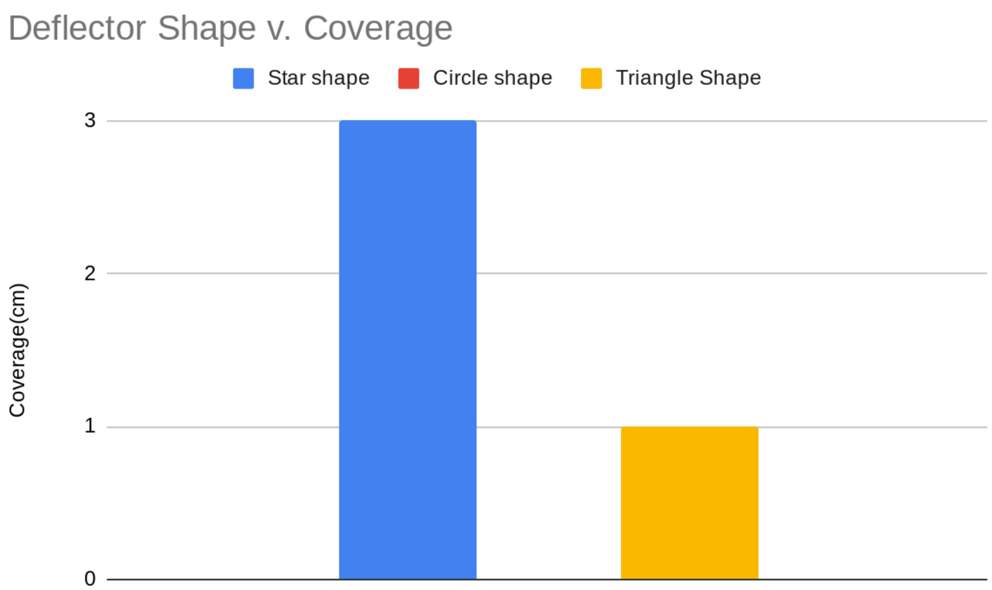

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| **Element I: Testing, Data Collection and Analysis** | | | | | | |
|---|---|---|---|---|---|---|
| | 5 | 4 | 3 | 2 | 1 | 0 |
| **1. Quantity of tests conducted.** The student conducts _____________tests for high priority requirements that are reasonable based on instructional context, or through physical or mathematical modeling. | *more than four* | *four* | *three* | *two* | *one* | *no* |

**Name:** ____________________________ **Date:** ________________ **Class:** ______________

| | | | | | | |
|---|---|---|---|---|---|---|
| **2. Importance of tests.** The student conducts tests for____________ requirements that are reasonable based on instructional context, or through physical or mathematical modeling. | *all high priority* | *most high priority* | *half of high priority* | *some low and high priority* | *only non priority* | *no* |
| **3. Comprehension of testing procedure.** The student demonstrates ______________ understanding of testing procedure, including the gathering and analysis of resultant data. | *considerable* | *ample* | *adequate* | *partial or overly general* | *minimal* | *a failure of* |
| **4. Detail of the design analysis.** The analysis of the effectiveness with which the design met stated goals includes ______________ explanation (and summary) of the data. | *a consistently detailed (generously* supported by pictures, graphs, charts and other visuals*)* | *a generally detailed (generally* supported by pictures, graphs, charts and other visuals*)* | *a somewhat detailed (at least somewhat* supported by pictures, graphs, charts and other visuals*)* | *a partial (partially complete and/or partially correct* and *minimally* supported by pictures, graphs, and other visuals*)* | *an attempted (may not be* supported by pictures, graphs, charts and other visuals*)* | *no* |